



TRACE AI MITIGATION INTELLIGENCE REPORT

PROJECT: Development of New Civil Enclave and associated works at Darbhanga Airport	Risk Classification: [MEDIUM]
Sector: Airport Passenger Terminal Building (NITB) Project ID: 75b48491-fcbf-42f2-9202-ad75773507ee	Composite Risk Score: 27.3/100 (Schedule: 44.8% Cost: 5.9%)

1. Executive Recommendation

Project 'Development of New Civil Enclave and associated works at Darbhanga Airport' (Airport Passenger Terminal Building (NITB)) in Darbhanga Airport & AIIMS is currently classified under the MEDIUM risk category (Risk Score: 27.3/100, Forecast Delay: ~4.0 months). At 65.0% physical completion with ~35.0% scope remaining in the Superstructure assembly and active corridor execution stage, immediate operational priority must focus on mobilize dedicated fast-track taskforce at darbhanga airport & aiims to clear critical-path bottleneck on milestone 'sit... Airport Authority of India [AAI] and Ministry of Civil Aviation should maintain active supervision to safeguard the revised completion milestone.

2. Key Risk Drivers (SHAP Feature Attribution)

Risk Factor	Impact	Evidence / Metric	Source
Budget spent 23.9% slower than progress	High Critical-Path Impact	SHAP feature attribution score of +0.268 on Development of New Civil Enclave and associated works at Darbhanga Airport	XGBoost SHAP Engine
108% of scheduled timeline elapsed	High Critical-Path Impact	SHAP feature attribution score of +0.205 on Development of New Civil Enclave and associated works at Darbhanga Airport	XGBoost SHAP Engine
Current physical progress reached: 65.0%	Moderate Risk Factor	SHAP feature attribution score of +0.042 on Development of New Civil Enclave and associated works at Darbhanga Airport	XGBoost SHAP Engine
Budget cost revision: +0.0%	Moderate Risk Factor	SHAP feature attribution score of +0.000 on Development of New Civil Enclave and associated works at Darbhanga Airport	XGBoost SHAP Engine

3. Why This Project Is At Risk (Root Cause Analysis)

Identified Risk: Milestone Execution Bottleneck Likely Root Cause: Critical path interlock on milestone 'Site Survey & Clearance (Completed)', impeding subsequent package handoffs in Darbhanga Airport & AIIMS. Observed Project Evidence: 3 delayed/pending milestone(s) identified with project progress recorded at 65.0%.
Identified Risk: Execution Velocity Lag Likely Root Cause: Physical throughput on large-span space frame roof truss erection, facade double glazing installation, and MEP primary ducting/piping risers is tracking behind the baseline master schedule, generating a cumulative ~4.0-month schedule overrun. Observed Project Evidence: Current certified progress is 65.0%, leaving ~35.0% scope to execute across Darbhanga Airport & AIIMS.
Identified Risk: Financial Expansion & Spend Alignment Likely Root Cause: Sanctioned outlay revised upward by Rs. 0.0 Cr (+0.0%), with fund utilization variance at -23.9%. Observed Project Evidence: Revised cost is Rs. 911.7 Cr (Original: Rs. 911.7 Cr; Spent: Rs. 374.5 Cr; Balance: Rs. 537.2 Cr).

4. Priority Mitigation Actions

Pri	Risk / Action	Responsible Role	Timeline	Expected Outcome & Escalation
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P1	[HIGH] Mobilize dedicated fast-track taskforce at Darbhanga Airport & AIIMS to clear critical-path bottleneck on milestone 'Site Survey & Clearance (Completed)', accelerating handover to subsequent package contractors. Evidence: Milestone tracking records 'Site Survey & Clearance (Completed)' as pending while overall physical delivery on Development of New Civil Enclave and associated works at Darbhanga Airport is at 65.0% with ~35.0% residual scope. Reason: Unblocking 'Site Survey & Clearance (Completed)' directly eliminates downstream dependency deadlocks and prevents further schedule slippage beyond the forecasted 4.0 months.	General Manager (Engineering-Civil) & Airport Director, Airports Authority of India [AAI]	Within 14 to 21 calendar days	Sustain certified physical execution velocity on Development of New Civil Enclave and associated works at Darbhanga Airport and prevent critical-path milestone divergence across remaining 35.0% scope. Escalate: Escalate critical package delays to Member (Planning), Airports Authority of India [AAI] for contractual performance review.
P2	[MEDIUM] Reconcile remaining capital outlay of Rs. 537.2 Cr with certified Measurement Book entries for Development of New Civil Enclave and associated works at Darbhanga Airport. Evidence: Cumulative spend is Rs. 374.5 Cr against sanctioned outlay of Rs. 911.7 Cr, leaving Rs. 537.2 Cr committed balance. Reason: Guarantees liquidity alignment and seamless vendor payments across active work packages in Darbhanga Airport & AIIMS.	Joint General Manager (Finance), Airports Authority of India [AAI]	Within 10 to 14 business days	Ensure 100% financial disbursement efficiency on completed physical scopes. Escalate: Escalate billing discrepancies to Chief Financial Officer / Finance Directorate, Airport Authority of India [AAI].
P3	[MEDIUM] Convene joint site coordination cell between Airport Authority of India [AAI], district administration, and utility departments at Darbhanga Airport & AIIMS to resolve Right-of-Way, utility shifting, and statutory clearance friction on Development of New Civil Enclave and associated works at Darbhanga Airport. Evidence: Corridor execution across Darbhanga Airport & AIIMS requires continuous encumbrance-free work fronts to support remaining 35.0% physical delivery. Reason: Eliminates external regulatory and land access friction that could compound the existing schedule trajectory.	General Manager (Engineering-Civil) & Airport Director, Airports Authority of India [AAI]	Next 21 to 30 calendar days	Achieve 100% unencumbered work-front availability across active packages in Darbhanga Airport & AIIMS. Escalate: Table encumbrance issues before District Magistrate / State Coordination Committee for emergency resolution.
P4	[HIGH] Deploy dedicated engineering audit team on Development of New Civil Enclave and associated works at Darbhanga Airport to address primary risk driver 'Budget spent 23.9% slower than progress', enforcing weekly monitoring protocols across Darbhanga Airport & AIIMS. Evidence: SHAP predictive model identified 'Budget spent 23.9% slower than progress' with feature attribution score of +0.268 on Development of New Civil Enclave and associated works at Darbhanga Airport. Reason: Targeting the dominant statistical risk contributor directly mitigates the algorithmic risk exposure computed by the predictive engine.	General Manager (Engineering-Civil) & Airport Director, Airports Authority of India [AAI]	14 to 21 Days	Neutralize risk contribution from 'Budget spent 23.9% slower than progress' and stabilize composite project reliability score. Escalate: Escalate critical package delays to Member (Planning), Airports Authority of India [AAI] for contractual performance review.

5. Continuous Monitoring Plan

Monitoring Indicator	Current Value	Target Value	Cadence	Accountable Role
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Physical Progress Velocity	65.0% certified completion	100% milestone synchronization across Darbhanga Airport & AIIMS	Monthly	General Manager (Engineering-Civil) & Airport Director, Airports Authority of India [AAI]
Expenditure vs Physical Balance	-23.9% variance gap (Spent: Rs. 374.5 Cr of Rs. 911.7 Cr)	Synchronized 0.0% variance	Monthly billing cycle	Joint General Manager (Finance), Airports Authority of India [AAI]
Schedule Delay Trajectory	4.0 months forecast delay (stable)	Zero net monthly slippage against revised target date	Monthly review	General Manager (Engineering-Civil) & Airport Director, Airports Authority of India [AAI]

6. Institutional Governance & Escalation Triggers

Trigger Event	Variance Threshold	Escalate To	Recommended Action
Schedule delay on Development of New Civil Enclave and associated works at Darbhanga Airport increases by > 1.5 months in consecutive review cycles	Monthly progress lag >= 3.0%	Escalate critical package delays to Member (Planning), Airports Authority of India [AAI] for contractual performance review.	Convene emergency project review for Development of New Civil Enclave and associated works at Darbhanga Airport and enforce contractual recovery milestones.
Financial burn rate on Development of New Civil Enclave and associated works at Darbhanga Airport exceeds verified physical progress by > 5.0%	Spend-progress variance > 5.0%	Principal Financial Adviser & Finance Committee, Ministry of Civil Aviation	Order independent third-party measurement book audit prior to releasing subsequent capital tranches.

AI Provenance & Audit Verification: Plan ID: MP-2026-3FA6949A | Version: v1 | Hash: 14379f360c71db1b | Primary Model: Qwen 2.5 (Dynamic Risk Reasoner) | Generated: 2026-09-05T08:37:35.600354